

Predicting Mental Health Trends

Data Acquisition and Per-Processing Report

Group 11
Aman Ostwal
Sai Pokuri
Sanjoli Sogani
Darshit Rai

February 9, 2025

1 Introduction

The purpose of this report is to describe the data acquisition process implemented in the Python script provided. The script retrieves data from Reddit using the Python Reddit API Wrapper (PRAW), processes it using the Natural Language Toolkit (NLTK), and saves the cleaned data in a CSV file. The primary focus is on extracting posts from subreddits related to mental health topics: **Anxiety, Depression, and ADHD**.

2 Data Sources

The dataset is acquired from **Reddit**, an online discussion platform.

2.1 Subreddits Collected

The script collects data from the following subreddits:

- `r/anxiety`
- `r/depression`
- `r/adhd`

2.2 Reddit Metadata Fields

Reddit provides structured metadata such as:

- **Post Title** – The title of the Reddit post.
- **Post Content (Selftext)** – The main body of the post.
- **Author** – The username of the person who posted.
- **Submission Timestamp** – The UTC timestamp of the post.
- **Score** – The net upvotes minus downvotes.

3 Acquisition Process

3.1 Downloading the Data

3.1.1 Is the Dataset Ready for Download?

- The dataset is not pre-existing for direct download; instead, it must be extracted dynamically from Reddit
- The script fetches posts in real-time using Reddit's API via **PRAW**.

3.1.2 Downloading Methodology

The script requires Reddit API credentials, including:

- `client_id`
- `client_secret`
- `user_agent`

These credentials allow access to Reddit's API. The script fetches the latest 10,000 posts per batch from each subreddit. Posts are fetched in chronological order until it reaches posts older than 10 days.

3.2 Implementation Code for Data Acquisition

Listing 1: Reddit Data Acquisition Script

```
import praw

reddit = praw.Reddit(
    client_id='your/_client/_id',
    client_secret='your_client_secret',
    user_agent='your_user_agent'
)

subreddit = reddit.subreddit("anxiety")
for post in subreddit.new(limit=10):
    print(post.title, post.selftext)
```

3.3 Need for Custom Code

Yes, custom Python code is required to fetch, process, and save the data.

The script uses:

- **PRAW**: To fetch data from Reddit.
- **NLTK**: To tokenize the text data.
- **Pandas**: To store the extracted data in a structured format.

3.4 Data Integration from Multiple Sources

The script fetches data from multiple subreddits and combines them into a single Pandas DataFrame. Each record includes a **Subreddit** column to indicate the source.

The data is cleaned by:

- Removing duplicates
- Converting timestamps to human-readable format
- Tokenizing text for NLP tasks

3.5 Integrating Code Example

```
import pandas as pd

data = []
subreddits = ["anxiety", "depression", "adhd"]
for subreddit_name in subreddits:
    subreddit = reddit.subreddit(subreddit_name)
    for post in subreddit.new(limit=10000):
        data.append({
            "Title": post.title,
            "Content": post.selftext,
            "Subreddit": subreddit_name,
            "Created_UTC": post.created_utc
        })

# Convert to DataFrame
df = pd.DataFrame(data)
```

4 Additional Processing Steps

4.1 Text Tokenization

The script tokenizes the `selftext` field to prepare for Natural Language Processing (NLP) tasks.

It uses NLTK's `word_tokenize()` function.

```
from nltk.tokenize import word_tokenize
import nltk
nltk.download('punkt')

# Example tokenization
post_text = "This is an example Reddit post. It needs tokenization."
tokens = word_tokenize(post_text)
print(tokens)
```

4.2 Deduplication

The script removes duplicate posts using Pandas' `drop_duplicates()` function.

```
df.drop_duplicates(subset=["Author", "Created_UTC", "Title"], inplace=True)
```

4.3 Export to CSV

The final processed data is saved as a CSV file for analysis.

```
output_file = "reddit_posts.csv"
df.to_csv(output_file, index=False)
print(f"Data saved to {output_file}")
```

4.4 Conclusion

- The script automates the data acquisition process from Reddit.
- It collects, processes, and saves structured data for further analysis.
- The integration of multiple subreddits ensures a comprehensive dataset.
- The process is scalable, allowing modification for additional subreddits or filters.

4.5 Issues

- The PRAW API only fetches data from the past 10–15 days and does not provide historical data.
- To obtain older data, access is required from either:
 - A Reddit Moderator of the subreddit.
 - Direct access from Reddit API, as Reddit has recently changed its terms and conditions.
- Without such access, obtaining historical data is not feasible with the current script.

5 Data Processing

5.1 Examining the Data and Understanding Its Properties

The dataset consists of Reddit posts with key attributes such as:

- **Author:** The username of the person who posted.
- **Title:** The title of the post.
- **Selftext:** The main body of the post.
- **Subreddit:** The subreddit where the post was made.

- **Created_UTC & Created_DateTime:** Timestamps indicating when the post was created.
- **Score:** The number of upvotes minus downvotes.
- **Tokenized_Selftext:** A tokenized version of the post's content.

To ensure accurate and meaningful analysis, we need to preprocess the data by:

- Removing missing values
- Removing inconsistencies
- Handling outliers
- Removing redundant data

5.2 Pre-Processing Requirements

Before analyzing the data, we need to preprocess the following issues:

- **Missing Data:** Some columns might have missing values that could impact analysis.
- **Duplicate Records:** Repeated posts can skew results.
- **Inconsistent Data Types:** Columns may have incorrect formats, making computations difficult.
- **Noise in Text Data:** Unnecessary special characters and extra spaces can interfere with text processing.
- **Outliers in Scores:** Extremely high or low scores can distort statistical insights.
- **Standardization Issues:** Subreddit names may have inconsistencies in capitalization and spacing.
- **Irrelevant Words (Stopwords):** Words like "the," "is," and "and" add little value to text analysis.

5.3 Data Preprocessing Steps

5.3.1 Handling Missing Data

Issue: Some key columns have missing values, leading to incomplete or biased insights.
Solution:

- Removed rows where **Author** and **Selftext** were missing.
- Dropped the **Selftext** column entirely and used upward fill to impute missing values in the **Author** column.

5.3.2 Duplicate Records

Issue: Duplicate posts can inflate the importance of certain data points. **Solution:** Removed duplicate rows to ensure each post is unique.

5.3.3 Inconsistent Data Types

Issue:

- `Created_UTC` should be an integer but might be stored as a string or float.
- `Created_DateTime` should be a proper datetime format.
- `Score` should be an integer, not a float or string.

Solution:

- Converted `Created_UTC` to integers.
- Converted `Created_DateTime` to datetime format.
- Converted `Score` to integers.

5.3.4 Noise in Text Data

Issue: Titles and post bodies contain:

- Uppercase and lowercase inconsistencies.
- Unnecessary special characters.
- Extra spaces.

Solution:

- Converted all text to lowercase.
- Removed special characters (e.g., `!@#$%^&*`).
- Trimmed extra spaces.

5.3.5 Handling Outliers in Scores

Issue: Some posts may have extremely high or low scores, which can skew insights.

Solution: Removed scores below the 1st percentile and above the 99th percentile to reduce outlier effects.

5.3.6 Standardizing Subreddit Names

Issue: Inconsistent capitalization or spacing in subreddit names may create duplicate categories. **Solution:** Converted all subreddit names to lowercase and stripped extra spaces.

5.3.7 Removing Stopwords

Issue: Common words like "the," "is," and "and" don't add value to text-based insights.

Solution: Removed stopwords from Tokenized_Selftext using the NLTK stopwords list.

6 Pseudo Code

```
BEGIN

SUBREDDITS = ["anxiety", "depression", "adhd"]
INITIALIZE Reddit API with CLIENT_ID, CLIENT_SECRET, USER_AGENT

POSTS_PER_BATCH = 10000
DATA = []
UNIQUE_POST_IDS = EMPTY_SET
ONE_YEAR_AGO_TIMESTAMP = CURRENT_UTC_TIME - (3650 days)

FOR EACH subreddit_name IN SUBREDDITS:
    SUBREDDIT = GET_SUBREDDIT(subreddit_name)
    FETCHED_POSTS = 0 // Counter for posts fetched
    LAST_POST_TIME = NULL // Track the oldest post fetched

    WHILE TRUE:
        TRY:
            // Fetch posts based on LAST_POST_TIME
            IF LAST_POST_TIME is not NULL:
                POSTS = FETCH_NEW_POSTS(SUBREDDIT, POSTS_PER_BATCH,
                    BEFORE=LAST_POST_TIME)
            ELSE:
                POSTS = FETCH_NEW_POSTS(SUBREDDIT, POSTS_PER_BATCH)

            POSTS_FETCHED = 0

            // Iterate through each fetched post
            FOR EACH POST IN POSTS:

                // If post is older than one year, stop fetching
                // for this subreddit
                IF POST.CREATED_UTC < ONE_YEAR_AGO_TIMESTAMP:
                    PRINT "Reached posts older than one year"
                    BREAK

                // Skip duplicate posts
                IF POST.ID IN UNIQUE_POST_IDS:
                    CONTINUE

                // Add post ID to unique tracker
                ADD POST.ID TO UNIQUE_POST_IDS
                POSTS_FETCHED += 1

                IF POST.SELFTEXT is not EMPTY:
                    TOKENS = TOKENIZE(POST.SELFTEXT)
                ELSE:
                    TOKENS = EMPTY_LIST
```

```

        CREATED_DATETIME = CONVERT_UTC_TO_DATETIME(POST.
            CREATED_UTC)

    // Store extracted post details
    APPEND TO DATA:
    {
        "Author": POST.AUTHOR.NAME (or NULL if no
            author),
        "Created_UTC": POST.CREATED_UTC,
        "Created_DateTime": CREATED_DATETIME,
        "Score": POST.SCORE,
        "Selftext": POST.SELFTEXT,
        "Tokenized_Selftext": TOKENS,
        "Subreddit": POST.SUBREDDIT.NAME,
        "Title": POST.TITLE
    }

    LAST_POST_TIME = POST.CREATED_UTC

    FETCHED_POSTS += POSTS_FETCHED
    PRINT "Fetched POSTS_FETCHED posts from subreddit_name,
        total unique posts: COUNT(UNIQUE_POST_IDS)"

    IF POSTS_FETCHED == 0:
        PRINT "No more posts available in subreddit_name."
        BREAK

    CATCH EXCEPTION E:
        PRINT "Error fetching posts from subreddit_name:", E
        WAIT 5 SECONDS BEFORE RETRYING

    WAIT 1 SECOND
    DF = CONVERT_TO_DATAFRAME(DATA)
    REMOVE_DUPLICATES(DF, COLUMNS=["Author", "Created_UTC", "Title"])

    OUTPUT_FILE = "reddit_posts_latest_31_January_2025.csv"
    SAVE_TO_CSV(DF, OUTPUT_FILE)

    PRINT "Data saved to OUTPUT_FILE"

END

```

Listing 2: Reddit Data Acquisition Pseudo Code

6.1 Sample Data

	Author	Created_UTC	Created_DateTime	Score	Selftext	Tokenized_Selftext	Subreddit	Title
0	Fair-Cucumber198	1.739063e+09	2025-02-09 00:56:22	1	I'm prescribed a bottle of 30 1mg xanax every ...	[I, 'm, prescribed, a, bottle, of, 30, 1mg, xa...	Anxiety	xanax makes me realize how bad my anxiety is
1	trilOnconsulting00	1.739062e+09	2025-02-09 00:52:37	1	I have a fun one to share in hopes it helps li...	[I, have, a, fun, one, to, share, in, hopes, li...	Anxiety	Anxiety, Authenticity and the 'Perfect Pringle'
2	FlowerInDawn	1.739062e+09	2025-02-09 00:51:50	1	I'm a 31M and I've been dealing with anxiety f...	[I, 'm, a, 31M, and, I, 've, been, dealing, wi...	Anxiety	Waking up Out of breath, Chest Pain and My Spo...
3	finja_unicorn	1.739062e+09	2025-02-09 00:48:52	1	!!! Trigger warning omotophobia!!!!nnnn have a...	[I, I, I, Trigger, warning, omotophobia, I, I, ...	Anxiety	I don't know what to do now
4	KoalaAdorable2418	1.739062e+09	2025-02-09 00:47:57	1	im so thirsty. im having an ood theme about Sc...	[im, so, thirsty, ..im, having, an, ood, them...	Anxiety	please help

6.2 Data Definition

Column Name	Data Type	Description
Author	String	The Reddit username of the post author. If the author is deleted, this may be None .
Created_UTC	Float	The UTC timestamp when the post was created, represented as a Unix epoch time (seconds since 1970-01-01).
Created_Date	String	The human-readable timestamp (YYYY-MM-DD HH:MM:SS) converted from Created_UTC .
Score	Integer	The Reddit post score, representing the net upvotes minus downvotes.
Selftext	String	The main text content of the Reddit post. This can be empty if the post does not contain body text.
Tokenized_Selftext	String	A list of tokenized words from the Selftext field.
Subreddit	String	The name of the subreddit where the post was published (e.g., "Anxiety").
Title	String	The title of the Reddit post, usually a short summary or main topic of discussion.

Table 1: Schema Definition for the Reddit Posts Dataset

Table of Contributions

The table below identifies contributors to various sections of this document.

No.	Section	Writing	Editing
1	Data Sources	Everyone	Everyone
2	Data Pre-Processing	Everyone	Everyone
3	Appendix	Everyone	Everyone